

Proposed Solution

A Comprehensive Measure of Well-Being | HDI Predictor

Solution Overview

Build a React + TypeScript web application that computes HDI using the official UNDP formula. Support 20 country presets. Deploy on Vercel for free global access.

Core Algorithm

- Health Index = $(\text{life_expectancy} - 20) / (85 - 20)$
- Education Index = $(\text{MYS}/15 + \text{EYS}/18) / 2$
- Income Index = $(\ln(\text{GNI}) - \ln(100)) / (\ln(75000) - \ln(100))$
- HDI = $(\text{Health} \times \text{Education} \times \text{Income})^{(1/3)}$

User Interface Design

- Dark themed background with animated world map
- Clean input form with country dropdown
- Color-coded tier result display
- Responsive layout for all screen sizes

Flask Backend (ML version)

- Train Linear Regression on HDI.csv (195 countries)
- Save model as model.pkl
- Flask routes: /, /Prediction, /predict
- Render result via Jinja2 templates